CURRICULUM VITAE

NAME: **Roger Brent**

BIRTH: December 28, 1955
Spartanburg, South Carolina

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EDUCATION AND TRAINING:

1973	B.A., Computer Science and Statistics, Minor in Mathematics, University of Southern Mississippi, Hattiesburg, Mississippi
1982	Ph.D., Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts, Laboratory of Mark Ptashne
1982–1985	Research Fellow, Department of Biochemistry and Molecular Biology, Harvard University, Cambridge, Massachusetts, Laboratory of Mark Ptashne

ACADEMIC APPOINTMENTS:

1985–1990	Assistant Professor, Department of Genetics, Harvard Medical School, Boston, Massachusetts
1985–1990	Assistant Molecular Biologist, Massachusetts General Hospital, Boston, Massachusetts
1985-1998	Member, Biological and Biomedical Sciences Graduate Program, Harvard Medical School, Boston, Massachusetts
1990–1998	Associate Professor, Department of Genetics, Harvard Medical School, Boston, Massachusetts
1990–1998	Associate Molecular Biologist, Massachusetts General Hospital, Boston, Massachusetts
1993–2000	Member, MD/PhD program, Harvard Medical School, Boston, Massachusetts
1997–2001	Associate Director, The Molecular Sciences Institute, Berkeley, California

1999–2011	Member, Program in Human Genetics, University of California, San Francisco, San Francisco, California
2000–2011	Adjunct Professor, Department of Biopharmaceutical Sciences, University of California, San Francisco, San Francisco, California
2001–2009	Research Director and President, The Molecular Sciences Institute, Berkeley, California
2002–2005	Director, Center for Genomic Experimentation and Computation, Berkeley California (TMSI)
2005–2009	Director, Center for Quantitative Genome Function, Berkeley, California (Center renamed in 2005)
2009–present	Professor, Division of Basic Sciences, Fred Hutchinson Cancer Research Center, Seattle, Washington
2009–present	Adjunct Professor, Division of Public Health Sciences, Fred Hutchinson Cancer Research Center, Seattle, Washington
2009–present	Member, Fred Hutchinson Cancer Research Center/ University of Washington Cancer Consortium, Seattle, Washington
2010-present	Affiliate Professor, Department of Genome Sciences, University of Washington, Seattle, Washington
2011-present	Member, Molecular and Cellular Biology Graduate Program, University of Washington/Fred Hutchinson Cancer Research Center, Seattle, Washington
2014-present	Affiliate Professor, Department of Bioengineering, University of Washington, Seattle, Washington

OTHER APPOINTMENTS:

1983	Visiting lecturer, Institute of Molecular Biology, Cuban Academy of Sciences, Havana, Cuba
1984–1993	Consultant, Bacterial and Yeast Genetics, Genetics Institute, Cambridge, Massachusetts
1988	Visiting lecturer, Department of Biochemistry, University of Arizona, Tucson, Arizona
1989	Visiting lecturer, Institute of Biological Research, Hungarian Academy of Sciences, Szeged, Hungary
1991–1993	Consultant, Developmental Biology, Genetics Institute, Cambridge, Massachusetts

1993–1994	Advisor to Scientific Board, Genetics Institute, Cambridge, Massachusetts
1994–1999	Member, Scientific Advisory Board, Genetics Institute, Cambridge, Massachusetts
1999–2001	Member, Scientific Advisory Board, Wyeth-Ayerst Research, Cambridge, Massachusetts and Madison, New Jersey
2000–2004	Chair, Scientific Advisory Board, Interomex Ltd., Vancouver, Canada
2000	Visiting Scientist, Institute of Pure and Applied Mathematics, University of California, Los Angeles, Los Angeles, California (summer)
2001	Visiting Lecturer, Institute of Theoretical Physics, University of California, Santa Barbara, Santa Barbara, California (summer)
2001 - 2003	Chair, Scientific Advisory Board, Plexus Vaccine Inc., San Diego, California
2001–2007	Founder, and chair of Scientific Advisory Board, Aptanomics SA, Lyon, France
2001–2009	Board of Trustees, Hybrid Vigor Institute, San Francisco, California
2003–2005	Member, Board of Directors, Siga Inc. (successor to Plexus), New York, New York
2021-Present	Founder and Scientific Advisor, LabLight AR, Inc.
2024-	Visiting Scientist, Center for AI, Security, and Technology, the RAND Corporation, Santa Monica, California and Washington, DC

PROFESSIONAL SOCIETIES:

1985–present	American Society for Microbiology
1985–present	American Association for the Advancement of Science
1990–present	American Society for Genetics
1999–present	American Society for Cell Biology
2006–present	Founder, member executive committee, International Society for Systems Biology

AWARDS AND HONORS:

1971–1973	University of Southern Mississippi, full university scholarship
1972–1973	University Honors College Stipend Award, University of Southern Mississippi

1973	National Merit Scholarship Finalist
1987–1991	Pew Scholar in the Biomedical Sciences
1991–1996	American Cancer Society Faculty Research Award
2001–2004	Senior Scholar, Ellison Foundation for Medical Research
2003	Jacob and Louise Gabbay Award in Biotechnology and Medicine (shared with Stanley Fields)
2010	Elected Fellow of the American Association for the Advancement of Science

INTERNATIONAL AND NATIONAL SERVICE:

Advisory activities:

1989–1990	*Ad hoc* member, microbiology and virology study section, American Cancer Society
1990–1992	Member, molecular biology and genetics study section, American Cancer Society
1993–1997	Member, Scientific Advisory Committee on Virology and Molecular Genetics, American Cancer Society
1995-2000	Founder and organizer "After The Genome" or ATG workshops, Santa Fe and Jackson Hole (I stopped these after ATG6 in 2000, when the completion of the human genome sequence was declared, and we had entered the after period)
1996–2001	Advisor, Unconventional Pathogen Countermeasures and Advanced Diagnostics, Defense Advanced Research Projects Agency (DARPA) Defense Sciences Office (DSO)
1996- 2009	Advisory work with NIH, including *ad hoc* attendance at study sections, chairing special study sections, and attendance and presentations at strategy meetings for NHGRI, NCI, the NIH Director, and NIDA.
1997–2009	Advisory work with Department of Energy (DOE), and Department of Defense (DOD) and other US government agencies.
1999 - 2009	Member, Biochem 2020, a US government group to advise on defense against biological attack.
1999–2004	Member, Functional Genomics Advisory Panel and Steering Committee, Wellcome Trust, London, England
1999–2007	Advisor, Program in Bioinformatics, University of California, Santa Cruz, Santa Cruz, California

2003–2007	Member, Scientific Advisory Board, *Saccharomyces* Genome Database, Stanford University, Palo Alto, California
2004–2007	Member, Strategy Committee, "Genes, Molecules, and Cells", Wellcome Trust, London, England
2004–2013	Member, Scientific Advisory Board, Pittsburgh Supercomputer Center, Pittsburgh, Pennsylvania
2004–2009	Selection committee, NIH Director's Pioneer Award
2005-2024	Founding member, "Biological Sciences Expert Group", a US government group to advise on defense against biological attack and other developments in the life sciences. My work concerned systems and genomic biology, new technologies, anticipation and management of the changes these developments will bring, and the nexus between this work and world efforts to counter naturally occurring infectious disease
2005-2024	Annual nominator and *ad hoc* evaluator for a major US prize. *Ad hoc* nominations for national and international prizes.
2006–2009	Founder and Executive Committee Member, International Society for Systems Biology
2006–2009	Chair, International Advisory Board, Biotechnology and Biological Sciences Research Council (BBSRC), Manchester Centre for Integrative Systems Biology, Manchester, England
2004 - 2009	Selection committee, NIH Director's Pioneer Award

Initiated or continued since coming to FHCC

Since joining FHCC, my most significant service has been national work with the US government on defense against biological attack and its nexus with naturally occurring infectious disease. This work is also international in that it helps protect the human population.

2009 – 2014	Executive Committee Member, International Society for Systems Biology
2009 - 2011	Chair, International Advisory Board, Biotechnology and Biological Sciences Research Council (BBSRC), Manchester Centre for Integrative Systems Biology, Manchester, England
2009 - 2014	Selection committee, NIH Director's Pioneer Award
2009 – 2013	Member, Scientific Advisory Board, Pittsburgh Supercomputer Center, Pittsburgh, Pennsylvania
2009 - 2023	Advisory work with NIH, including *ad hoc* attendance at study sections, attendance and presentations at strategy meetings for NCI.

2009 – 2024	Advisory work with Department of Energy (DOE), Department of Defense (DOD) and other US government agencies.
2009 - 2014	Member, Biochem 2020, a US government group to advise on defense against biological attack. Group disbanded in 2014.
2009 - 2014	Founder and Executive Committee Member, International Society for Systems Biology
2009 - 2024	Founding Member, Biological Sciences Expert Group, a standing US government group to advise on defense against biological attack and other developments in the life sciences. My recent work has largely concerned the impact of systems and genomic biology, new technologies, anticipation and management of the changes these developments will bring, and the nexus between this work and world efforts to counter naturally occurring infectious disease.
2009 -2024	Annual nominator and *ad hoc* evaluator for a major US prize. *Ad hoc* nominations for other national and international prizes.
2009 - 2011	Science Advisor, The Hastings Center, "Study on synthetic biology and nonphysical moral concerns"
2011 - 2013	Science Advisor, The Hastings Center, follow-on study, "The ethical issues of synthetic biology"
2014 - 2018	Scientific Advisory Board, International Society for Systems Biology
2015 - 2016	Member, Biodefense Working Group, Presidential Council of Advisors in Science and Technology, White House
2016	*Ad hoc* advisor, "Future Challenges in Cell Biology", Wellcome Trust.
2023-2024	Member, Brotman-Baty Institute, Seattle

Editorial boards:

1997–2001	Editor, Molecular Cell
1987–present	Current Protocols in Molecular Biology (cot'd since coming to FHCC)
1990 - present	Oversight editorial review of daughter Current Protocols journals (continued since coming to FHCC)

Manuscript Reviews (2009-present)

Nature, Science, Cell, Nature Genetics, Cell Systems, PloS journals, Current Protocols journals

Invited talks (2009-present)

More than 100 talks and seminars, about half in international meetings.

Other National and International Service (2009 - present)

2009- present	Evaluations for academic promotions, 14
2009- present	Nominations and arguments in favor of prize recipients, 11
2009- 2016	Approximately twice-monthly input into thought, analysis, and decisions by US government agencies and White House (EOP), contact with US House and Senate staff.
2017- 2020	Approximately monthly input into work by USG agencies and/ or contact with Congress, no contact with EOP 2017-2020.
2021-2024	Resumed more regular contact with US government including EOP
2024-2026	Contact with USG.

LOCAL SERVICE:

Prior to joining Fred Hutchinson Cancer Center

1985–2009	Supervised 4 undergraduates, 2 post-Baccs, 5 PhD students, 16 postdocs, and 3 visiting scientists
1985–1990	Graduate admissions committee, Harvard Medical School (HMS) Biological and Biomedical Sciences (BBS) graduate program
1986–1988	Instructor, Genetics 208, Harvard Medical School One of three faculty members. 20 graduate students. 15 hours preparation and contact time/week (for ~5 weeks/semester)
1989–1994	Director, Genetics 205a and Genetics 205b, "Control of cell identity and organismic pattern," Harvard Medical School. Course director, and one of four faculty. 6–10 graduate students each semester. 10 hours preparation and contact time, when teaching (~10 weeks/year), 1 hour administrative work each week otherwise (20 weeks/year)
1990–1991	Organizing committee, Harvard Medical School BBS retreat
1991–1998	Seminar committee, Massachusetts General Hospital (MGH) Department of Molecular Biology
1993–1999	Admissions committee, Harvard Medical School MD/ PhD program
1993–1994	Lectures in basic molecular biology for support staff in MGH DMB. 6-8 support staff (technicians, secretaries, etc.) 1 contact hour/ two weeks, for ~24 weeks /year (12 sessions)

1994	Instructor, Micro 213, "Social issues in biology," Spring 1994, Harvard Medical School. One of four faculty. 15 graduate students. 20 hours preparation and contact time/week for (2 weeks/ year)
1994–1995	MGH Subcommittee on the Review of Research Proposals
1994–1998	HMS Steering committee, qualifying exams, BBS program
1994–1998	Instructor, Microbiology 230, "Analysis of the biological literature," the big MIT "730"-type course mandated by the new BBS core curriculum. ~12 Preparation hours/week + 3 contact hours/week for the whole semester (11 weeks)
1995–2000	Member, Committee on Higher Degrees in Biophysics, Harvard University, Cambridge, Massachusetts
1995–1998	MGH/ Partners committee on research
2002–2009	Genomic curriculum development, University of California, San Francisco and University of California, Berkeley
2003–2005	Taught Letters and Sciences 126, "Genomics and Citizenship," an upper-level pan-UC Berkeley course. Course co-sponsored by NIH/ NHGRI and co-taught with Paul Rabinow, Professor of Anthropology, University of California, Berkeley
2005–2008	Scientific Advisory Board, University of California, Berkeley Synthetic Biology Engineering Research Center

Local service since joining FHCC (2009-present)

In terms of consistent commitment and having made a difference, the most important local service work I do is probably my participation in the partnership for the cancer research minority program that FHCC runs with New Mexico State University (NMSU) in Las Cruces

2009-present	Supervised 2 visiting senior scientists, 2 staff scientists, 3 graduate students (these with NMSU), 1 postdoctoral fellow with NMSU, 4 postdoctoral fellows here, 3 post-Bacc students, and 4 pre-Bacc students
2010-2011	Mentor and coach for FHCC junior faculty member
2011-2014	Started and led Center for Biological Futures, a pilot project at FHCC to bring about better thinking about the impacts that increases in biological technology are having on human affairs.
2011-2014	Started daughter effort at UW, Biological Futures in a Globalized world, led by Alison Wylie (Philosophy) and Kathleen Woodward (Simpson Center).
2009- 2020	Significant work in support of FHCC-NMSU Underrepresented Minority Program. Mentor and coworker for NMSU junior faculty members.

2014, 2015, 2016	Weintraub award evaluations committee
2011, 2015 -2017	Graduate admissions evaluations, Molecular and Cellular Biology Program
2016-2020	Mentor and coach for FHCC junior faculty member
2022	Outreach in concert with small foundation to five HBCUs in US Southeast

TRAINEES (since 2009)

Sabbatical/ Visiting Scientist

Laura Boucheron, PhD (2019-2024), Computer vision/ machine learning

Marsha Maxwell, PhD (2021 -2024), Human factors/ UI/UX research.

James Ashley (2021-2024), Augmented Reality design and development

Dominic Travis, DVM (2023-2024), Flows of animal products and pathogens through ecology.

Postdoctoral

Kohtaro Tanaka (2020 - 2023), Director, Genome editing, SalioGen Therapeutics, Waltham, Massachusetts

Alexander Mendenhall (2011 - 2016), Associate Professor, Department of Pathology, University of Washington

Meg Stalcup (2009 - 2014), Assistant Professor, Sociological and Anthropological Studies, University of Ottawa

Gaymon Bennett (2011 - 2014), Assistant Professor of Religion, Science, and Technology, Arizona State University

Guillermo Acosta (2011 -2014), multiple times weekly contacts and visits here, joint with Houston lab, NMSU), Founder, Biofuel Engine Components, Berino, New Mexico

Ippolytos Kalofonos, (2011 - 2014), Assistant Professor, Department of Psychiatry, West Los Angeles VA Medical Center and UCLA Center for Social Medicine and Humanities

Graduate

Patrick Jenkins (2009 -2015, multiple times weekly contacts and visits here). Chemical engineering, NMSU, now Research Engineer, Polysciences Inc., Indianapolis, Indiana

Fredrick Crawford (2012-2014, multiple times weekly contacts and visits here). Chemical engineering NMSU, now grade 11-12 Mathematics Instructor, Miami-Dade County Public School District, Miami, Florida

Gabriella LaBazzo (2019-2020), School of Public Health, University of Washington, now WSDOH.

Swetha Sanapoori (2020), School of Public Health, University of Washington

Post-Bacc

Derek Britain (2010-2014), PhD program in Biophysics, UCSF, now Scientist at 4:59 NewCo, San Francisco

Charlie Hahn (2011 - 2013), now PhD program in Anthropology, UW

James Adam Redfield (2009 - 2011), PhD, Department of Religious Studies, Stanford University, now Assistant Professor of Biblical and Talmudic Literature, Department of Theological Studies, Saint Louis University.

Karrington T. Ogans (2019-2023), in lab.

Gabriella LaBazzo (2019-2021), now WSDOH

Gaea Turman (2020-2022), now Allen Institute for Cell Science.

Undergraduate

Katie Ulrich (2011-2012), Anthropology, Haverford College, now PhD Student, Department of Anthropology, Rice University

Hani Mahmoud (2012), Bioengineering, University of Washington, now Fulbright Scholar, Dasman Diabetes Institute, Kuwait City, Kuwait

Newsha Farahani (2012-2014), Bioengineering, University of Washington, now manufacturing engineer at pH Science Holdings, Lynwood, Washington

Sylvia Stellmacher (2012), Physiology, University of Washington, now UNICEF, Seattle, Washington

John Munar (2020-2021), Computer Science, Seattle University

Emma Zucati (2020-2021), Computer Science, Seattle University

Baszin Belhu (2020), Computer Science, University of Washington

Michael Bremer (2020-2021), Physics, University of Washington

J. Anthony Mein (2021), Computer Science, University of Washington, now LabLight AR

Cindy Ou (2021), Public Health, University of Washington

Nik Stires (2021), Computer Science, Seattle University, now LabLight AR

Hannah Merritt (2021), Neuroscience, University of Washington

Aiko Torkelson (2022), Biology, Seattle University, now Fred Hutch

SELECTED PUBLICATIONS AND NON PRINT MATERIALS (WORK AT FHCC BEGINS ON PAGE 24)

PUBLICATIONS BEFORE COMING TO FHCC

1. Brent, R., and Ptashne, M. The wild-type *lexA* gene product represses its own promoter. Proc. Natl. Acad. Sci. 1980, 77, 1932-1938.

Molecular cloning of LexA and demonstration that LexA represses a gene induced in the SOS response (recA) and also itself. 36 years later, in order to bring about better control in eukaryotes, we are trying to reproduce the same autoregulatory feedback control in eukaryotes.

2. Brent, R., and Ptashne, M. Mechanism of action of the *lexA* gene product. Proc. Natl. Acad. Sci. 1981, 78, 4204-4208.

Overproduction of LexA, purification, DMS and DNAse mapping of binding sites in regulatory regions of genes in vitro*, and careful measurement of differential affinity for different genes under conditions mimicing the E. coli cytosol. At the time we thought these differential affinities regulated timing of induction of SOS response genes, now we also believe that LexA's autorepression provides closed loop feedback control of a type not found in eukaryotes.*

3. Brent, R. Regulation and autoregulation by *lexA* protein. Biochimie 1982, 64, 565-569.

4. Kenyon, C., Brent, R., Ptashne, M., and Walker, G. Regulation of damage inducible genes in *Escherichia coli*. J. Mol. Biol. 1982, 610, 445-457.

The regulatory regions of Cynthia Kenyon's din genes, identified as LexA repressed phage Mu dlac genome insertions, are bound by LexA protein in vitro.

5. Ebina, Y., Takahara, Y., Kishi, F., Nakazawa, A., and Brent, R. LexA protein is a repressor of the Colecin E1 gene. J. Biol. Chem. 1983, 248, 13258-13261.

Defines LexA operators upstream of ColE1 as the superstrongest ones in the universe. If there were four LexA monomers in the cell and a single ColE1 promoter, that promoter would be completely repressed.

6. Brent, R. Regulation of the *E. coli* SOS response by the *lexA* gene product. In: Friedburg, E. C. , Bridges, B. A, eds. UCLA symposia on molecular and cellular biology. New Series 1983; 11, 361-368.

7. Brent, R., and Ptashne, M. A bacterial repressor protein or a yeast transcriptional terminator can block upstream activation of a yeast gene. Nature 1984, 312, 612-615.

LexA enters eukaryotic nucleus, recognizes DNA sites despite chromatin, and represses transcription. Argues against one idea for transcription activation (unwinding) and sets up modular activator experiments.

8. Brent, R., and Ptashne, M. A eukaryotic transcriptional activator bearing the DNA specificity of a prokaryotic repressor. Cell 1985, 43, 729-736.

Establishes modular structure of eukaryotic transcription regulatory proteins. Kicks off short lived era of establishing modules by nailing protein domains to DNA. Contributes to two-hybrid methods. Contributes to start of age of less-thoughtful, less structure based protein engineering.

9. Brent R. Repression of transcription in yeast. Cell 1985; 42: 3-4.

Lays out difference in mechanism of artificial repression by prokaryotic proteins in eukaryotes vs. active repression native to eukaryotes.

10. Silver, P. A. , Brent, R., and Ptashne, M. DNA binding is not sufficient for nuclear localization of regulatory proteins in *Saccharomyces cerevisiae*. Mol. Cell. Biol. 1986, 6, 4763-4766.

Demonstration of Brent and Ptashne 1984, that the ~48 Kd LexA dimer and the ~220 Kd LexA-Gal4 dimer gets into the nucleus even though that lack nuclear localization sequences and are not specifically localized.

11. Brent. R. How to run a very small factory. Cell 1986, 47m 329-330

12. Brent, R. Current Protocols in Molecular Biology, Chapter 1, "E. coli, plasmids and phages." 1987-2016. Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K. Struhl) John Wiley and Sons, New York.

I wrote most of this chapter initially and keep maintaining it.

13. Brent, R. Current Protocols in Molecular Biology, chapter 12, "DNA-protein interactions" (backup editor, main editor, Robert Kingston). 1987-2016. Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K. Struhl) John Wiley and Sons, New York.

I did some writing for and continue to perform editorial work for this chapter.

14. Lech, K., Anderson, K., and Brent, R. DNA-bound Fos proteins activate transcription in yeast. Cell 1988, 53, 179-184.

First demonstration of transcription regulation by DNA bound nuclear oncoprotein complexes.

15. Samson, M.-L., Jackson-Grusby, L., and Brent, R. Gene activation and DNA binding by *Drosophila* Ubx and abd-A proteins. Cell 1989, 57, 1045-1052.

Separates DNA binding by homeodomain from transcription regulation by rest of proteins. Argues that negative regulation requires additional Drosophila proteins.

16. Hanes, S. D., and Brent, R. DNA specificity of the Bicoid activator protein is determined by the homeodomain recognition helix residue 9. Cell 1989, 57, 1275-1283.

Shows genetically in yeast that Bicoid has different specificity than Ubx and ftz, and that the specificity is due to a particular residue in the Bicoid recognition helix. Work not well appreciated by Drosophila biologists working on this problem

17. Kamens, J., Richardson, P., Mosialos, G., Brent, R., and Gilmore, T. Oncogenic transformation by vRel requires an amino-terminal activation domain. Mol. Cell. Biol. 1990, 6, 2840-2847.

18. Brent, R. Current Protocols in Molecular Biology, chapter 16, "Protein expression.". 1990-2016. Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K. Struhl) Greene Publishing, Brooklyn, New York.

I wrote substantial pieces of this chapter, and maintain it by performing substantial editorial work for it.

19. Hanes, S. D., and Brent, R. A genetic model for the interaction of the homeodomain recognition helix with DNA. Science 1991, 251, 426-430.

Shows that published X-ray structure, over interpreted, had the bicoid DNA recognition helix oriented backwards to the way it must be in vivo. Paper took > 12 months to publish due in part to opposition from structural biologists.

20. Kamens, J., and Brent, R. Native Rel and Dorsal proteins recognize distinct DNA sequences. New Biologist 1991, 10, 1005-1013

21. Golemis, E., and Brent, R. Fused protein domains inhibit DNA binding by LexA. Mol. Cell. Biol. 1992, 12, 3006-3014.

Set up genetic means to demonstrate that potential LexA fusion bait proteins would function in two-hybrid assays.

22. Brent, R., Finley, R. Golemis, E., Jessen, T., Xu, C. W., Mendelsohn, A., Cohen, B., Colas, P., Cohen, R. Lok, L. Brent lab www server, xanadu.mgh.harvard.edu, 1993-1998, www.molsci.org, 1998-2009. Now at Russ Finley's lab http://proteome.wayne.edu/THPL.html

We began distributing strains, plasmids and working cDNA libraries for our lab "interaction trap" two-hybrid method before publication, first via a Gopher server and then via the "world wide web". In retrospect, my decision not to publish our lab's work on the problem, including our experiments to build working methods based on Fields and Song 1989 until everything worked well, was a mistake. However, we shipped reagents to more than 5,000 labs worldwide and many reported success.

23. Zervos, A., Gyuris, J., and Brent R. Mxi1, a protein that specifically interacts with Max to bind Myc/Max recognition sites. Cell. 1993, 72, 223-232.

Co-published with Max from Eisenman lab. Zervos invented the word "bait".

24. Gyuris, J., Golemis, E., Chertkov, H., and Brent, R. Cdi1, a human tyrosine phosphatase that associates with Cdk2 during G1. Cell. 1993, 75, 791-803.

Methods paper describing the method, mainly the work of Jeno Gyuris and Erica Golemis.

25. Edelhoff, S., Ayer, D. E., Zervos, A.S., Steingrimsson, E., Jenkins, N.A., Copeland, N.G., Eisenman, R.N., Brent, R., and Disteche, C. Mapping of two genes encoding members of a distinct subfamily of Max interacting proteins: Mad to human chromosome 2 and mouse

chromosome 6, and Mxi1 to human chromosome 10 and mouse chromosome 19. Oncogene 1994, 9, 665-668

26. Hanes, S., Riddehough, G., Ish-Horowicz, D., and Brent, R. Specific DNA recognition and intersite spacing are critical for action of the Bicoid morphogen in *Drosophila*. Mol. and Cell Biol. 1994, 14, 3364-3375

27. Cuomo, C., Kirch, S., Gyuris, J., Brent, R., and Oettinger, M. A. Rch1, a protein that specifically interacts with RAG-1 and affects V(D)J recombination. Proc. Natl. Acad. Sci. USA. 1994, 91, 6156-6160

28. Hanes, S. D. and Brent, R. Backwards Bicoid? Structure, 1994, 2, 894.

Wherein we note that the structural biologists had gotten it backwards, and still had not acknowledged this.

29. Parroush, Z., Finley, R., Brent, R., and Ish-Horowicz, D. Interactions between Groucho and bHLH proteins mediate transcription repression during Drosophila neurogenesis, segmentation, and sex-determination. Cell, 1994, 79, 805-816

30. Finley, R. L., and Brent, R. Binary and ternary interactions between Drosophila cell cycle regulators. Proc. Natl. Acad. Sci. USA, 1994, 91, 12980-12984

This described "interaction mating", allowing scale-up of two-hybrid and making it "genomic". It laid out the use of pre-made "panels of baits", comprised of different bait proteins, to give insight into biological function. It laid out the idea that patterns of protein interactions might give insight into function even without knowledge of the interacting proteins. It launched the "bait return project", under which we collected >1,000 working LexA fusion bait proteins that we used in panels to guess function of new proteins.

31. Golemis, E. A., Gyuris, J. G., and Brent, R. Interaction traps / two hybrid systems to identify interacting proteins. 1994 chapter 13, unit 14 in Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K. Struhl) John Wiley and Sons, New York (1987-2016).

32. Mendelsohn, A, and Brent, R. Biotechnology applications of interaction trap/ two-hybrid systems. Current Opinion in Biotechnology 1994, 5, 482-486.

33. Lee, J. W., Choi, H., S., Gyuris, J., Brent, R., and Moore, D. D. Two classes of proteins dependent on either the presence or absence of thyroid hormone for interaction with the thyroid hormone receptor. Molecular Endocrinology, 1995, 9, 243-254.

34. Khazak, V., Sadhale, P., Woychik, N. A., Brent, R., and Golemis, E. A. Human RNA polymerase II subunit hsRPB7 functions in yeast and influences stress survival and cell morphology. Mol. Biol. Cell, 1995, 6, 759-775.

35. Su, L-K., Burrell, M., Hill, D.E., Gyuris, J., Brent, R., Wiltshire, R., Trent, J., Vogelstein, B., and Kinzler, K.W. APC binds to the novel protein, EB1. Cancer Research, 1995, 55, 2972-2977

Vogelstein lab had used our reagents before publication to show that MDM2 bound p53. I had asked that my name not appear on that paper. This is a follow on paper.

36. Estojak, J., Brent, R., and Golemis, E. Correlation of two hybrid affinity data with *in vitro* measurements. Mol. Cell. Biol. 1995, 15, 5820-5829

This is the key careful study that shows how to interpolate, using data from defined affinity protein-protein interactions and different reporter genes with different numbers and affinities of LexA operators in our interaction trap two-hybrid system, to estimate the affinity of new protein-protein interactions. The high-water mark of quantification use of this in vivo *technology.*

37. Reymond, A. and Brent, R. p16 from human tumors shows diminished interaction with Cdk4. Oncogene, 1995, 11, 1173-1178.

The paper shows that p16 interacts with Cdk4 and Cdk6, and that allelic forms of p16 from cancer prone families do not. It also shows one p16 variant that interacts normally with Cdk6 but not with Cdk4. That shows that the impact of p16-G101W on public health is not likely to be due to an effect on its interactions with Cdk6.

38. Steingrimsson, E., Sawadogo, M., Gilbert, D., Zervos, A.S., Brent, R., Blanar, M.A., Fisher, D.E., Copeland, N.G., and Jenkins, N.A. Murine chromosomal location of five bHLH-zip transcription factor genes. Genomics, 1995, 28, 179-183.

39. Zervos, A., Faccio, L., Gatto, J., Kyriakis, J., and Brent, R. Mxi2, a mitogen-activated protein kinase that recognizes and phosphorylates Max protein. Proc. Natl. Acad. Sci. USA. 1995, 92, 10531-10534

Although we did not get first to print, this was the first isolation of p38 MAPK (we would now call this isoform α, and the ortholog with the same function is of course S. cerevisiae Hog1)

40. Clemens, Karen E., Brent, R., Gyuris, J., and Munger, K. Characterization of protein/protein interaction domains of the human papillomavirus E7 oncoprotein. Virology, 1995 214, 289-293

41. Colas, P., Cohen, B., Jessen, T., Grishina, I., McCoy, J., and Brent, R. Genetic selection of peptide aptamers that recognize and inhibit Cyclin-dependent kinase 2. Nature, 1996, 380, 548-550.

The first peptide aptamers, conformationally constrained variable regions of random sequence selected from libraries, some of which bind tightly to most intracellular targets. Peptide aptamers have continued to develop (see for example Reverdatto et al. 2015, Peptide Aptamers, Development and Applications, Curr Top Med. Chem, 15, 1082-1101).

42. Finley, R., and Brent, R. The interaction trap. In DNA cloning - expression systems: a practical approach. (ed D. Glover and B. D. Hanes) 1996 Oxford University Press, Oxford, England.

43. Finley, R., Thomas, B.J., Zipursky, L., and Brent, R. Isolation of *Drosophila* Cyclin D, whose expression in the morphogenetic furrow precedes entry into S. Proc. Natl. Acad. Sci. USA, 1996, 93, 3011-3015

44. Thomas, B. J., Zavitz, K. H., Dong, X., Lane, M. E., Weigman, K., Finley, R., Brent, R., Lehner, C. F., and Zipursky, S. L. *roughex* downregulates G2 cyclins in G1. Genes and Development, 1997, 10, 1289-1298.

We found that in a screen with a panel of baits against Cyclin E. That same screen picked up one of our previous Drosophila Cdk interactors and characterized it. Cdi4 was later identified as decapo, and we were never able to publish the original work (Finley, R. L. Jr., Cohen, B, and Brent, R. Drosophila Cdi4 is a p21/p27/p57-like cyclin-dependent kinase inhibitor with specificity for cyclin E complexes, but it lives online at http://proteome.wayne.edu/Cdi4ms.html)

45. Meroni, G., Reymond, A., Borsani, G. Tanigami, A., Tonlorenzi, R., Lo Nigro, C., Messali, S., Zollo, M., Ledbetter, D. H., Brent, R., Ballabio, A., and Carrozzo, R. Rox, a novel bHLHZip protein expressed in quiescent cells that heterodimerizes with Max, binds a non-canonical E box, and acts as a transcriptional repressor. EMBO Journal, 1997, 16, 2892-2906.

46. Xu, C. W., Mendelsohn, A., and Brent, R. Cells that register logical relationships among proteins. Proc. Natl. Acad. Sci. USA, 1997, 94, 12473-12478.

Describes development of two-hybrid variant and protein aptamers to allow analysis of different partner relationships between closely related allelic protein variants. By performing logical operations on the phenotypes, we make inferences about the topology and order of protein interactions of these proteins in complexes. This was the high water mark of my lab's efforts to try to find genetic means to probe protein complex formation, and for our attention to digital logic circuitry inside cells. A more lasting contribution was this first isolation of peptide aptamers specific for a disease form allelic states, here, activated human hRas but not wild-type hRas, laying the groundwork for peptide aptamers and proteins like them to be used as research tools and therapeutic agents.

47. Golemis, E. and Brent R., 1997. The interaction trap. In "The Two Hybrid System", P. Bartel and S. Fields, editors, Oxford University Press, London.

48. Finley, R, and Brent R., 1997. Two-hybrid analysis of genetic networks. In "The Two Hybrid System", P. Bartel and S. Fields, editors, Oxford University Press, London.

49. Brent, R. Looms to weave genomic nets. Nature Genetics, 1997, 16, 216-217.

50. Brent, R. Raising the next generation of scientific pioneers. Journal of NIH Research, 1997. 9, 37-38.

51. Brent, R., and Finley, R. L., Jr. Studying gene and allele function with two-hybrid methods. Ann. Rev. Genetics, 1997, 31, 663-704.

A idea-rich review with an thoughtful former postdoc about how particular findings about protein interactions allow formation or testing of specific hypothesis of the function of different allelic forms of proteins, with suggested application to the human genome. I believe a large number of these ideas still merit testing.

52. Marti, F., Xu, C. W., Selvakuman, A., Brent, R., Dupont, B., and King, P. D. LCK phosphorylated human killer cell inhibitory receptors recruit and activate phosphatidylinositol 3-kinase, Proc. Natl. Acad. Sci. USA, 1998, 95, 11810-11815

53. Cohen, B., Colas, P, and Brent, R. A novel cell cycle inhibitor isolated from a combinatorial library, Proc. Natl. Acad. Sci. USA, 1998, 95, 14272-14277

First demonstration of use in mammals: Peptide aptamers selected in yeast specific to Cdk2 selected in yeast inhibit progression of Saos-2 cells through G1.

54. Brent, R. Current Protocols in Molecular Biology, chapter 20, Protein interactions (editor, backup editors Kevin Struhl and Robert Kingston). 1998-2016. Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K.Struhl) John Wiley and Sons, New York.

I did substantial writing for this chapter.

55. Krainc, D., Bai, G., Okamoto, S., Carles, M., Kusiak, J. W., Brent, R., Lipton, S. A. Synergistic activation of the N-methyl-D-aspartate receptor subunit 1 promoter by myocyte enhancer factor 2C and Sp1. J. Biol. Chem. 1998 273, 26218-26224

56. Mendelsohn, A. R., and Brent, R. Postgenomic protein analysis: the next bend in the river. Nature Biotechnology, 1998, 16, 520-521.

57. Colas, P., and Brent, R. The impact of two-hybrid and related methods on biotechnology. Trends in Biotechnology, 1998, 16, 355-363.

58. Brent, R. Functional genomics: learning to think about gene expression data. Current Biology, 1999, 9, 338-341.

59. Geyer, R., Colman-Lerner, A., and Brent, R. "Mutagenesis" by peptide aptamers identifies genetic network members and pathway connections. Proc. Natl. Acad. Sci. USA, 1999, 96, 8567-8572

A "mutant hunt" using peptide aptamers to screw up multiple points in a signaling pathway and find their targets. Found Cbk1, which we would later show kinases Mob1 in mother daughter asymmetry (these two are at the heart of the Hippo planar cell polarity system in metazoans).

60. Fabbrizio, E., Le Cam, L., Polanowska, J., Kaczorek, M., Lamb, N., Brent, R., and Sardet, C. Inhibition of mammalian cell proliferation by genetically selected peptide aptamers that functionally antagonize E2F activity, Oncogene, 1999, 18, 4357-4363

Strong inhibition in mammalian cells.

61. Brent. R., and Carlson, R. Double stranded DNA arrays: next steps in the surface campaign, Nature Biotechnology, 1999, 17, 536-537.

62. Mendelsohn, A. R., and Brent, R. Protein interaction methods: toward an endgame. Science, 1999, 284, 1948-1950.

63. Brent, R. Current Protocols in Molecular Biology, chapter 22, Nucleic acid arrays, 1999. Current Protocols in Molecular Biology (ed F. M. Ausubel, R. Brent, D. D. Moore, J. Seidman, J. A. Smith, and K. Struhl (1987-2016)) John Wiley and Sons, New York.

I did some writing for this chapter.

64. Raj, L., Hanes, S. D., Appel, L. F., Badam, E. Fernandez, M., Finley, R. L. Jr., Brent, R., and Weir, M. (2000) Targeted localized degradation of Paired protein in *Drosophila* development. Current Biology, 2000, 10, 1265-1272

Use of derivatized protein aptamers to degrade targeted proteins.

65. Meroni, G., Cairo, S., Merla, G., Messali, S., Brent, R., Ballabio A, Reymond A. (2000) Mlx, a new Max-like bHLHZip family member: the center stage of a novel transcription factor regulatory pathway? Oncogene, 19, 3266-3277

66. Colas, P., Cohen, B., Ferrigno, P., Silver, P. A., and Brent, R. (2000) Targeted modification and transportation of cellular proteins. Proc. Natl. Acad. Sci. USA, 97, 13720-13725

Use of derivatized peptide aptamers to drag targeted proteins to different subcellular locations.

67. Brent, R. (2000) Genomic Biology, Cell, 100, 169-183.

This review in the "millennium issue" helped people think about what they meant by "genomic biology" and (I hope) a more thoughtful consideration of what might be meant by "systems biology". My particular goal was to make research to gain predictive quantitative understanding of biological system behaviors a legitimate goal of systems biology.

68. Brent, R. and Kodadek, T. (2000) Interaction, assembly and processing: Life on an interface. Current Opinion in Chemical Biology, 4, 3-15.

69. Brent, R. (2000) Using artificial reagents to dissect cellular genetic networks, in Landweber, L. F. and Winfree, E., eds. Evolution as Computation, Springer-Verlag, Berlin.

70. Geyer, R., and Brent, R. (2000). Selection of genetic agents from random peptide aptamer expression libraries. Methods Enzymol. 328, 171-208.

71. Colman-Lerner, A. and Brent, R. (2000). Using peptide aptamers to analyse the proteome. Trends in Cell Biology, New Technologies in the life sciences, a Trends Guide, Supplement, 12, 56-60

A good short review of the promise of more widespread development of peptide aptamers as research tools. Peptide aptamers and molecules based on the same logic had some application as research agents and more than 10 are in human clinical trials. DOI-ed and available at FHCC archives.

72. Colman-Lerner, A. Chin, T. and Brent, R. (2001). Yeast Cbk1 and Mob2 proteins kinase activate distinct daughter-specific genetic programs that induce asymmetric cell fates. Cell, 107, 739-750

Isolation of genes specifically expressed in daughter cells but not their mothers. In cerevisiae, the previously unknown genes are DSE (for Daughter Specific Expression) 1-4. Exploration of the expression of one such daughter specific gene, CTS1, revealed how the kinase Cbk1, and the associating Mob1 localize a transcription activator, Ace2, to the daughter nucleus. Not surprisingly, the metazoan analogs of Cbk1 and Mob1, respectively Warts/LATS1 and 2, and Mats/MOBKL1A and B, for Mob1, seem to do the same thing to give rise to asymmetric cell division as part of "core cassette" at the heart of the hippo pathway.

73. Endy, D. and Brent, R. (2001). Modeling cellular behavior. Nature, 409, 391-395

Models and their limitations, still largely true.

74. Mendelsohn, A. Hamer, J. Zhaoyuan, B.W. Brent, R. (2002). Cyclin D3 activates Caspase 2, connecting cell proliferation with cell death. Proc. Natl. Acad. Sci. USA, 99, 6871-6876

Powerful means using fluorescent proteins to select mutant proteins deficient in interaction with a specific partner but not with others.

75. Ptashne, M. Gill, G, and Brent, R. (2003). Modularity of eukaryotic transcriptional activators. 2003. In "Great experiments", www.ergito.com

76. Brent, R. (2004). Intuition and innumeracy. Cell Biol. Educ., 3, 88-90

77. Brent, R. Building an artificial regulatory system to understand a natural one. Cell, 2004, 116 (2 Suppl) S73-S76

Tries to summarize context of the problem at the time.

78. Brent, R. (2004) A partnership between biology and engineering. Nature Biotechnology, 22, 1211-1214.

This is a careful article that continues to have influence on the field.

79. Burbulis, I., Yamaguchi, K., Carlson, R., and Brent, R. (2005) Using protein-DNA chimeras to detect and count small numbers of molecules. Nature Methods 2, 31-37

Tadpoles, peptide aptamers with DNA tails covalently joined by playing a trick with intein excision. Use to count bound antibody-like molecules with the sensitivity of PCR. Demonstration of beautiful statistical means to make such counts maximally accurate.

80. Lok, L. and Brent, R. Automatic generation of cellular reaction networks with Moleculizer 1.0. Nature Biotechnology, 2005, 23, 131-136.

Automatic generation of models when the number of possible permutations of particular protein complexes exceeds the ability of humans to enumerate them or code them. We developed this approach, which is now widely used.

81. Brent, R. and Lok, L. (2005) A fishing buddy for hypothesis generators. Science 2005, 308, 504-506

Review/commentary on Sachs et al. (Nolan, Lauffenberg) and use of Judeah Pearl/ Bayesian network methods to infer causal relationships from high-throughput flow cytometry data.

82. Brent. R. In the valley of the shadow of death (2003-2006) MIT Synthetic Biology Archive http://hdl.handle.net/1721.1/34914, DOI: 1721.1/34914, https://dspace.mit.edu/bitstream/1721.1/34914/1/Valley2006.pdf

Strategic overview and proposed overarching 21st century for defense against biological attack. From 2003 and revised 2005-2006. Parts of this analysis figure in baseline strategic analysis used by US government and its tenets are, I believe, universally acknowledged.

83. Colman-Lerner, A., Gordon, A., Serra, E., Chin, T., Resnekov, O., Endy, D., Pesce, G. and Brent, R. (2005) Regulated cell-to-cell variation in a cell fate decision system. Nature. 437, 699-706

Presents some basic methods for yeast single cell work (cells growing in 96 well cover-slip-bottomed plates, glued to bottom of well with concanavalin A, imaging software, etc. now used worldwide). Shows how to dissect and quantify contributions to cell to cell variation into variation in signal through pathway and variation in global gene expression power, G, and how to break the second into a separately quantifiable weakly heritable long lived component and a shorter lived stochastic or noise component. The basis for all our subsequent work on cell to cell variation in yeast and C. elegans.

84. Brent, R., Bruck, J. (2006) 2020 computing: can computers help to explain biology? Nature, 440, 416-417

Has a meditation on the difference between function selected by evolution and function as understood by computer scientists. I plan to revisit this article in 2025.

85. Brent, R. Power and responsibility. (2006) MIT Synthetic Biology Archive http://hdl.handle.net/1721.1/34913, DOI: 1721.1/34913, https://dspace.mit.edu/bitstream/1721.1/34913/1/PowerandResponsibility.pdf

86. Thomson, T., Benjamin, K.R, Endy, D. and Brent, R. 2006-2016 Yeast pheromone response wiki http://yeastpheromonemodel.org

This is maintained by Thomson, then a graduate student with Drew Endy, and it still gets visited occasionally.

87. Gordon, A., Colman-Lerner, A., Chin, T.E., Benjamin, K.R., Brent, R. (2007) Single-cell quantification of molecules and rates using open source microscope based cytometry. Nature Methods, 4, 175-181

Describes all our software and tricks to use imperfect optics and maturing and decaying fluorophores to permit precise quantification of proteins from fluorescent signals in yeast. Was used in Colman-Lerner et al. 2005, and all our subsequent work. Describes in detail Cell-ID, the workhorse image quantification program.

88. Burbulis, I., Yamaguchi, K., Yu, R. C., Resnekov, O., and Brent, R. (2007) Using a "near-universal" protein-DNA chimera to detect and count small numbers of antibodies. Nature Methods 4, 1011-1013

Precise use of newly developed anti-Fc tadpoles to count bound antibodies and thus accurately quantify rare proteins.

89. Soergel, D., Choi, K., Thompson, T., George, B., Morgan-Linial, R., Brent, R., and Endy, D. MONOD: a collaborative tool for manipulating biological knowledge (2008) MIT Synthetic Biology Archive https://dspace.mit.edu/handle/1721.1/41867

A knowledge sharing setup to accompany creation of models, ahead of its time.

90. Herrgård M. J., Swainston, N., Dobson, P., Dunn, W. B., Yalçin Arga, K., Arvas, N.,Blüthgen, N., Borger,, S., Costenoble, R., Heinemann, M., Hucka,, M., Li, P., Liebermeister, W., Mo, M. L., Oliveira, A. P. Petranovic D., Pettifer, S., Simeonidis, E., 88. Smallbone, K., Spasic, I., Weichart, D., Brent, R., Broomhead, D. S., Westerhoff, H. V., Kırdar, B., Penttila, M., Klipp, Palson, B. Ø., Sauer. U., Oliver. S. G., Mendes, P. Neilsen, J., and Kell, D. B. (2008). A

consensus yeast metabolic network obtained from a community approach to systems biology. Nature Biotechnol. 26, 1155-1160

91. Yu, R. C., Resnekov, O., Abola, A. P., Andrews, S. S, Benjamin, K. R., Bruck, J., Burbulis, I. E., Colman-Lerner, A., Endy, D., Gordon, A., Holl, M., Lok, L., Pesce, C. G., Serra, E., Smith, R. D., Thomson, T. M., Tsong, A. E., and Brent R. The Alpha project, a model system for systems biology (2008) IET Syst. Biol. 2, 222-233.

92. Yu, R., Gordon, A., Colman-Lerner, A., Benjamin, K.R., Pincus, D., Serra, E., Holl, M., Brent, R. (2008) Negative feedback optimizes information transmission in a cell signaling system. Nature, 456, 755-761

Establishes Dose Response Alignment and its role in optimizing information transmission. Demonstrates one important feedback and from that suggests that system control depends on feedback, an idea we now believe to be false.

SELECTED PUBLICATIONS SINCE COMING TO FHCC

(Excludes most contributions to Current Protocols journals)

93. Brent, R. (2009) Cell signaling: what is the signal and what information does it carry? FEBS Lett. 583, 4019-4024

This review from a Nobel symposium still summarizes key questions for signal control. These include "the counting problem", how the system generates the information about the fraction of receptors bound, and "DoRA", Dose-Response Alignment, how the downstream responses mirror fraction receptor occupied on the cell surface.

94. Andrews, S. S., Addy, N. J., Brent, R., and Arkin, A. P. (2010). Detailed simulations of cell biology with Smoldyn 2.1. PLoS Comput Biol. 6(3): e1000705.

This is the first instance of the stochastic Smoldyn simulator to incorporate automatic generation of different combinations of proteins in complexes to generate all the reaction networks. This brings into stochastic simulation the idea of the "Moleculizer" work I did with Larry Lok in 2005.

95. Thomson, T. M., Benjamin, K. R., Thomson, T. M., Love, T., Yu, R., Gordon, A., Colman-Lerner, A., Endy, D. and Brent, R. (2011) Scaffold number in yeast signaling system sets tradeoff between system output and dynamic range. Proc. Natl. Acad. Sci. USA, 108, 20265-20270.

This is the publication of many person-years of careful work measuring the abundance of 17 key protein components of the PRS. It shows that the previous "whole genome" measurements of protein abundance were frequently inaccurate and that these inaccuracies have consequences for system performance. These numbers are key for all our subsequent quantitative work.

96. Pincus, D., Ryan, C. J., Smith, R. D., Brent, R., and Resnekov, O. (2013) Assigning quantitative function to post-translational modifications reveals multiple sites of phosphorylation that tune yeast pheromone signaling output. PLoS One. 8(3):e56544

This was the output of many person-years mapping phosphorylation sites on PRS proteins that only appear after system induction.

97. Flanigon, J., Kamali-Moghaddam, M., Burbulis, I., Annink, C., Steffen, M., Oeth, P., Brent, R., van den Boom, D., Landegren, U., and Cantor, C. (2013) Multiplex protein detection with DNA readout via mass spectrometry. Nature Biotechnology 30, 153-158.

Adds to our 2007 anti-antibody tadpole work the greatly increased specificity of proximity ligation assays, albeit at the cost of accuracy in quantification, while allowing fast readout via mass spec. I contributed intellectually to this.

98. Ventura, A. C., Bush, A., Vasen, G., Goldín, M. A., Burkinshaw, B., Bhattacharjee, N., Folch, A., Brent, R., Chernomoretz, A., and Colman-Lerner A. (2014) Utilization of extracellular information before ligand-receptor binding reaches equilibrium expands and shifts the input dynamic range. Proc. Natl. Acad. Sci. USA. 111(37): e3860-3869

How the pheromone response system and other systems with slow ligand binding to receptor can activate allows the systems to achieve a correct level of activation even before equilibrium receptor binding is reached. This mech probably operates in the glutamanergic AMPA receptors.

99. Sands, B., Jenkins, P., Peria, W. J., Naivar, M., Houston, J. P., and Brent, R. (2014) Measuring and sorting cell populations expressing isospectral fluorescent proteins with different fluorescence lifetimes. PLoS ONE. 9(10):e109940

First paper from collaboration with Jessica Perea Houston at New Mexico State. We make lifetime-shifted fluorescent proteins and show that we can sort cells of the same color but different lifetimes on Jessica's machine.

100. Mendenhall, A. R., Tedesco, P. M., Johnson, T. E., and Brent, R. (2015). Single cell quantification of reporter gene expression in live adult *Caenorhabditis elegans* reveals reproducible cell-specific expression patterns and underlying biological variation. PLoS ONE 10(5): e0124289.

This work, learning how to make low-noise reporter genes, and to get other sources of noise from measurement low enough to reveal natural underlying variation, is critical, as it laid the technical groundwork for all our subsequent quantitative work in worms.

101. Sands, B. and Brent, R. (2016). Overview of post Cohen-Boyer methods for single segment cloning and for multisegment DNA assembly. Curr. Protoc. Mol. Biol. 113:3.26.1-3.26.20.

The idea for publishing this was to give context for our publication of the Sands yeast recombination and ligation chain reaction (LCR) DNA assembly methods, which we hope to write up late 2016.

102. Mendenhall, A., Driscoll, M., and Brent, R. (2016). Using measures of single-cell physiology and physiological state to understand organismic aging. Aging Cell. 15(1):4-13.

The ideas in this essay seem to be trickling out gradually.

103. Cao, R., Jenkins, P. Peria, W., Sands, B., Naivar, M., Brent, R, and Houston, J. P. (2016). Phasor plotting with time-resolved flow cytometry. Optics Express. 24(13), 14596-14607

Second paper from our New Mexico collaboration. Most of the work extending phasor analysis from microscopy to flow cytometry came from me and from Bill Peria, a physics staff scientist in my lab.

104. Andrews, S., Peria, W., Yu. R,. C., Colman-Lerner, A., and Brent, R. (2016). Feedback and push-pull mechanisms can align signaling system outputs with inputs. Cell Systems 3 (5): 444-455.e2

Shows that most kinds of closed loop feedback control can't result in Dose-Response alignment and thus maximize information transmission, that another, inferior kind of control, Push-Pull, can do so, that feedback control can bring about alignment if it includes a component called a "comparator adjuster" that provides a closed-loop, and that, although closed loop control does not seem to operate in eukaryotic cell signaling systems, suggests a way to build it.

105. Mendenhall, M., Crane, M. M., Tedesco, P. M. Johnson, T. E. and Brent, R. Brent (2017) *Caenorhabditis elegans* genes affecting interindividual variation in life-span biomarker gene expression. J. Gerontol. A Biol. Sci. Med. Sci. 72, glw349.

First measurement of variation in lifespan biomarker gene as a phenotype and demonstration of two systems that affect variation.

106. Mendenhall, A. R., Leiser, S. Sutphin, B., Tedesco, P. M., Kaeberlein, M., Johnson, T. E, and Brent, R. (2017). Environmental canalization of lifespan and gene expression in *C. elegans*. *Journal of Gerontology A Biol Sci Med Sci, 2017, Vol. 00, No. 00, 1–5 doi:10.1093/gerona/glx017*

Temperature decreases lifespan, variation in lifespan and also variation in gene expression of lifespan biomarkers. Possibly due to low level induction of heat shock proteins which decrease various kinds of stochastic variation in operation of intracellular systems. An intriguing correlation, only.

107. Pesce, C. G., Zdraljevic, S., Peria, W. J., Bush, A., Repetto, M. V., Rockwell, D., Yu, R. C., Colman-Lerner, A., and Brent R. (2018). Single-cell profiling screen identifies microtubule-dependent reduction of variability in signaling. Mol Syst Biol. 2018 Apr 4;14(4):e7390. doi: 10.15252/msb.20167390.

Technically sweet work on a difficult problem, where does the noise come from. Some good tricks, including use of epistasis bypass experiments to get beyond the steps in the PRS where noise is introduced.

108. Andrews, S. S., Brent, R, and Balázsi, G (2018) Signaling Systems: Transferring information without distortion. eLife 2018;7:e41894 doi: 10.7554/eLife.41894

109. Brent, R. and Boucheron, L. (2018) Building and training neural networks to predict microscopic images. *Nature Methods.* 15(11):868-870, doi: 10.1038/s41592-018-0194-9

110. Elbing, K. F., and Brent, R. (2018) Recipes and tools for culture of *Escherichia coli*. *Current Protocols in Molecular Biology* 9:e83. doi: 10.1002/cpmb.83

111. Elbing, K. F. and Brent, R. (2018) Growth of *E. coli* in liquid medium. *Current Protocols in Molecular Biology.* 9:e81. doi: 10.1002/cpmb.81

112. Elbing, K. F. and Brent, R. (2018). Growth of *E. coli* on solid media. *Current Protocols in Molecular Biology* 10:e82. doi: 10.1002/cpmb.82

113. Crane, M. M., Sands, B., Battaglia, C., Johnson, T. E.., Yun, S., Kaeberlein, M., Brent, R., and Mendenhall., A (2019) *In vivo* measurements reveal a single 5'-intron is sufficient to increase protein expression level in *Caenorhabditis elegans*. Sci Rep. 2019 Jun 24;9(1):9192. doi: 10.1038/s41598-019-45517-0

114. Burnaevskiy, N., Sands, B, Yun, S., Tedesco. P. M., Johnson, T. E., Kaeberlein, M., Brent, R., and Mendenhall, A. (2019) Chaperone biomarkers of lifespan and penetrance track the dosages of many other proteins. Nat Commun. 2019 Dec 16;10(1):5725. doi: 10.1038/s41467-019-13664-7. PMID: 31844058

115. Azizoğlu, A., Brent, R., and Rudolf, F. (2021). A precisely-adjustable variation-suppressed transcriptional controller to enable genetic discovery. Elife. 2021 Aug 3;10:e69549. doi: 10.7554/eLife.69549. PMID: 34342575 PMCID: PMC8421071 DOI: 10.7554/eLife.69549

The "expression clamp". Uses precepts lifted from control theory to suppress variation around given setpoints. Expression clamped alleles define a new kind of allele.

116. Azizoğlu, A., Loureiro C., Venetz, J, and Brent, R. (2023) Autorepression-Based Conditional Gene Expression System in Yeast for Variation-Suppressed Control of Protein Dosage. Current Protocols 3, e647 https://doi.org/10.1002/cpz1.647

Protocols article for Azizoğlu et al. 2021, incorporating experience from other researchers using it and understanding of the reason expression clamp works in all growth media.

117. Andrews, S. S., and R. Brent. (2025) Andrews, S. S., and Brent, R. (2025) Individual Yeast Cells Signal at Different Levels but Each with Good Precision. *BioRχiv 90686.* Royal Society Open Science. 2025;12:241025. doi: 10.1098/rsos.241025. 5.

118. Azizoğlu, A., Bijman, E., Stelling, J. and Brent R. (2024). Azizoğlu, A., Bijman, E. Y., Stelling, J., and Brent, R. (2025) Evolverator: An Engineered In Cellulo Yeast System to Drive Rapid Continuous Evolution of Proteins. bioRxiv 2024.09.01.610536. doi: 10.1101/2024.09.01.610536 and bioRxiv 2024.09.01.610536v2

AI x Bio PUBLICATIONS FROM RAND ERA

119. Brent, R., McKelvey, T. G., Jr., and Matheny, J. (2024) The New Bioweapons: How Synthetic Biology Could Destabilize the World. Foreign Affairs, Vol. 103, No. 5, pp. 148-159. https://www.foreignaffairs.com/world/new-bioweapons-covid-biology

120. Brent, R., and McKelvey, T. G., Jr. (2025) Contemporary AI Foundation Models Increase Biological Weapons Risk. arXiv 2506.13798. RAND Corporation, PE-A3853-1. doi: 10.48550/arXiv.2506.

121. Williams, A. E., Del Castello, B., Lee, J., Roberts, D., Tarangelo, J. P., Atanda, J., Colman-Lerner, A., Gerold, J. and Brent, R. (2026) Developing a Risk-Scoring Tool for Artificial Intelligence–Enabled Biological Design. A Method to Assess the Risks of Using Artificial Intelligence to Modify Select Viral Capabilities. RAND Corporation, RR-A4490-1. As of March 25, 2026: https://www.rand.org/pubs/research_reports/RRA4490-1.html

122. Barkan, C., Rodriguez, C., Chowdhury, S., Zhang, L.-A., and Brent, R. (2026). Restricting AI Agent Use of Biological Tools: exploring the feasibility of software barriers. RAND Corporation, RR-A5000-1. https://doi.org/10.7249/RRA5000-1

OTHER WRITING UNLIKELY TO BE PUBLISHED

123. Ashley, J., Brekelmans, J., Peria, W. Brown, A, and Brent, R (2024) Information-rich, high-spatial-accuracy coordinate systems to support Augmented Reality-guided work. *In early prep.*

Our Augmented Reality System. Allen Brown, Jr. is a mathematical logician formerly at Microsoft Research. Brown extended an existing process calculus that supports this work.

Subject of appropriate patent applications, may not ever be published.

124. Tanaka, K, Peria, W., and Colman-Lerner, A., and Brent, R. (2026) Expression clamped control of gene expression in vertebrate cells. *In early prep.*

Tunable expression clamp controllers for vertebrate cells. May not ever be published.

125. Peria, W., Clark A. M., Boucheron, L. and Brent, R. (2026) CellNet: training data for machine learning in histopathology and cell biology. *Was in early preparation for Nature Scientific Data.*

Progress in AI has rendered this work irrelevant.

126. Seewald, A., Mendenhall, A. R., Boucheron, L., and Brent, R. (2024) CeNTMOOT, a machine learning tool to support human quantification of gene expression from single cells in tissues.

A human finds the nucleus, and then a random forest of decision trained trees decides the nuclear perimeter. We showed, contrary to some current AI doctrine, that some of our highly trained tress are smarter at this than others. Boucheron is an image analysis/ machine learning person at NMSU. Progress in AI has rendered this work irrelevant.

C) OTHER PUBLICATIONS AT FHCC FROM PEOPLE IN MY LAB

(As a policy, I do not insist on authorship on papers from my lab unless my intellectual contribution was strong)

Publications by Steve Andrews

2009-2020 Staff Scientist, Brent Lab, Division of Basic Sciences, FHCC

1. Andrews, S. S. (2009) Accurate particle-based simulation of adsorption, desorption and partial transmission. Phys Biol. 6(4):046015. doi: 10.1088/1478-3975/6/4/046015.

Describes more additions to the physical processes Smoldyn can represent.

2. Gladkov, D. V., Alberts, S., D'Souza, R. M., and Andrews, S. S. (2011) Accelerating the Smoldyn spatial stochastic biochemical reaction network simulator using GPUs. In Proceedings

of the 19th High Performance Computing Symposia, pp. 151-158. Society for Computer Simulation International, 2011.

The GPUs came from NVIDIA graphics cards used for video gameplay.

3. Andrews, S. S. (2012) Spatial and stochastic cellular modeling with the Smoldyn simulator. Methods Mol Biol. 2012;804:519-42. PMID: 22144170

Methods paper setting out more new abilities for Smoldyn, significantly including automatic reaction network generation ala Lok and Brent 2005.

4. Bouuaert, C., Lipkow, K, Andrews, S. S. Liu, D, and Chalmers, R. (2013). The autoregulation of a eukaryotic DNA transposon. Elife. 2:e00668. doi: 10.7554/eLife.00668. PMID: 23795293

5. Schmidt, H. G., Sewitz, S., Andrews, S. S., and Lipkow, K. (2014) An integrated model of transcription factor diffusion shows the importance of intersegmental transfer and quaternary protein structure for target site finding. PLoS One. 2014 Oct 21;9(10):e108575. PMID: 25333780

6. Andrews, S. (2014). Methods for modeling cytoskeletal and DNA filaments. Physical Biology 11(1):011001, PMID: 24476634

Adds flexible linear molecules to the physical entities that this simulator can represent.

7. Emiola, A., George, J. and Andrews, S. S. (2015) A Complete Pathway Model for Lipid A Biosynthesis in Escherichia coli. PLoS One. 10(4): e0121216.

8. Robinson, M., Andrews, S. S., and Erban R. (2015) Multiscale reaction-diffusion simulations with Smoldyn. Bioinformatics. 2015 Jul 15;31(14): 2406-2408.

Adds a simplified model that, when used, makes Smoldyn go faster.

9. Emiola, A., Andrews, S. S., Heller, C., and George J. (2016). Crosstalk between the lipopolysaccharide and phospholipid pathways during outer membrane biogenesis in Escherichia coli. Proc. Natl. Acad. Sci. USA. 113(11): 3108-3113. PMID: 26929331

10. Andrew, S. S. and Rutherford, S. (2016). A method and on-line tool for maximum likelihood calibration of immunoblots and other measurements that are quantified in batches. (2016). PLoS One. 11(2):e0149575.

I did have some conceptual input into this work.

11. Andrews, S. S (2017) Smoldyn: particle-based simulation with rule-based modeling, improved molecular interaction and a library interface. Bioinformatics. 2017 Mar 1;33(5):710-717. doi: 10.1093/bioinformatics/btw700.

12. Andrews, S. S. (2019) Rule-Based Modeling Using Wildcards in the Smoldyn Simulator. Methods Mol Biol. 2019;1945:179-202. doi: 10.1007/978-1-4939-9102-0_8.

Publications by research fellows Meg Stalcup and Gaymon Bennett, who came to my lab to learn biology and biological technology, and the intersection with security

Meg Stalcup

(Brent lab, 2009-2014) Cultural anthropologist, now Associate Professor, Sociological and Anthropological Studies, University of Ottawa

Articles In Refereed Journals

1. Stalcup, M. and Verguet S. (2011) Global health and the demands of the day. Health, Culture & Society, 1(1), 28-44.

2. Stalcup, M, and Verguet, S. (2012), Mothers and Children: Designing research toward integrated care for both. Health, Culture & Society, 3(1), 160-171.

3. Verguet, S., Stalcup, M., and Walsh, J. (2013) Where to deploy Pre-Exposure Prophylaxis (PrEP) in sub-Saharan Africa?" Sexually Transmitted Infections, 89(8), 628-634.

4. Stalcup, M. (2013) Patrimônio etnobotânico no Brasil: a feira livre. Revista Cadernos do CEOM, 26(38), 131-153.

Chapters In Edited Volumes (peer-reviewed)

5. Stalcup, M. (2011) Lab Notes: Write-up of an experiment in collaborative anthropology. in The Accompaniment: Assembling the Contemporary, Paul Rabinow, Chicago, University of Chicago Press, pp. 132-139.

6. Stalcup, M. (2013) Interpol and the Emergence of Global Policing and Contemporary Governance: The Anthropology of Police in Practice, William Garriott, Ed., New York, Palgrave MacMillan, pp. 231-261.

7. Stalcup, M. (2015) Policing Uncertainty: On Suspicious Activity Reporting. In Modes of Uncertainty: Policy and experience of uncertainty in the contemporary world, Limor Samimian-Darash and Paul Rabinow, Eds., Chicago, Chicago University Press, pp. 69-87.

Magazine Articles

8. Stalcup, M. and and Craze, J. (2011), How We Train Our Cops To Fear Islam. Washington Monthly, March/April.

Book Reviews

9. Stalcup, M. (2013). Spiritual Biologicals. Review of On The Internationalization of Ayahuasca, Labate and Jungaberle (2011); and A Hallucinogenic Tea, Laced with Controversy: Ayahuasca in the Amazon and the United States, Dobkin de Rios and Rumrill (2008), BioSocieties, 8(2), 234–238, June.

10. Stalcup, M. (2015). Review of Nicolas Langlitz's Neuropsychedelia: The Revival of Hallucinogen Research Since the Decade of the Brain, Nicolas Langlitz (2012), Somatosphere, April 2015

Gaymon Bennett

(Brent lab, 2011-2014) Cultural anthropologist and religious philosopher, now Assistant Professor of Religion, Science, and Technology, Arizona State University.

Books

1. Bennett, G. and Rabinow, P. Designing Human Practices: An Experiment with Synthetic Biology. University of Chicago Press, Chicago, 2012

Work done before Bennett did before he came with me to FHCC, but inflected by discussions with me.

2. Bennett, G. Technicians of Human Dignity: Bodies, Souls, and the Making of Intrinsic Worth. Just Ideas Series, Fordham University Press, 2015

3. Bennett, G., ed. Science, Reason, Modernity: Readings for an Anthropology of the Contemporary. Edited with Anthony Stavrianakis and Lyle Fernley. Fordham University Press, 2015.

4. Bennett, G., ed. The Ethics of Biotechnology. Volume editor in The Library of Bioethics, Wendell Wallach, series ed. Ashgate Publishers, 2016.

Articles and book chapters.

5. Bennett, G. "Science and Fabrications: On Synthetic Biology," (with Anthony Stavrianakis) Biosocieties (2014) 9, 219-223.

6. Bennett, G. "Nature Natured and Nature Denatured," Hastings Center Report Nov-Dec 2014, ppS38-9.

7. Bennett, G. "The Malicious and the Uncertain: Biosecurity, Ethics, and the Arts of Living," in Darash and Rabinow eds. Modes of Uncertainty: Policy and Experience of Uncertainty in the Contemporary World. University of Chicago Press, forthcoming, 2015.

8. Bennett, G.. "The Ethics of Biotechnology," introduction to Gaymon Bennett ed. The Ethics of Biotechnology. In The Library of Bioethics, Wendell Wallach, series ed. Ashgate Publishers, forthcoming 2016.

9. Bennett, G.. "The Moral Economy of Biotechnical Facility," Journal of Responsible Innovation, Volume 2, Issue 1, January 2015, pages 128-132.

10. Bennett, G. "The Ethics of Ubiquitous Computing in Healthcare," (with Clark Miller, Benjamin Hurlbut, and Heather Ross) in Mary Eshaghian-Wilner, ed. Pervasive Computing in Medicine: From Electronic Health Records to Nanomedicine. John Wiley & Sons, forthcoming 2016.

11. Bennett, G. "Contemporary Equipment for Anthropological Problems of Modern Sciences," in Science, Reason, Modernity: Readings for an Anthropology of the Contemporary, Fordham University Press, 2015.

D. Patents (lists issued and pending US patents. Numbers 1-10 refer to main inventions, which often split into multiple issued patents. Those US patents had cognate international patents. Those have now expired)

Before coming to FHCC

1. Brent, R. and Ptashne, M. "Regulation of eukaryotic gene expression." 1989. US Patent 4,833,080.

2. Zervos, A., and Brent, R. "Max interacting peptides and related molecules and methods" 1996 and 1998. US Patents 5,512,473, 5,780,262, and 5,852,169.

3. Gyuris, J., Golemis, E., Zervos, A., and Brent, R. 1996 "Interaction trap system for isolating novel peptides." US Patent 5,580,736.

4. Lech, K. F., Anderson, C., Golemis, E. A., and Brent, R. 1996 "Assays for inhibitors of Myc oncoprotein." US Patent 5,580,721.

5. Finley, R., and Brent, R. "Interaction trap systems for the analysis of protein networks." 1997. US Patent 5,695,941.

6. Gyuris, J., Golemis, E., and Brent, R. "Cdi1 and related peptides." 1998. US Patent 5,786,169.

7. Jessen, T., McCoy, J. and Brent, R. 1999 "Thioredoxin interaction trap for selecting useful peptides." US Patent 6,004,746.

8. Xu, C. W., Mendelsohn, A., Lok, L., and Brent, R. 2000 and 2005 "Detection systems for registering protein interactions and functional relationships." US Patents 6,132,953 and 6,858,382

9. Brent, R., McCoy, J.M., and Jessen, T. H. 2001 "Interaction trap systems for detecting protein interactions." US Patent 6,242,183

10. Carlson, R., Burbulis, I., and Brent, R. "Spectroscopic protein fingerprint system." 2005. US Patent 6,669,685

Since 2009

11. Brent, R. Augmented Reality Physical Properties Engine 2015, FSP 1129, US Patent 10,467,534 granted 11 November 2019, *name given by USPTO Augmented Reality Procedural System.*

12. Brent, R. Augmented Reality Procedural System, 2015, FSP1128, US Patent 10,706,626, Granted July 2020. *Same USPTO title as US 10,467,534, covers quite different ground.*

13. Brent, R. Procedural Language and Content Generation Environment for use of Augmented Reality/Mixed Reality systems to support laboratory and related operations. FSP1306, 2018, US Patent 10,963,750, granted 11 March 2021.

14. Brent, R. Augmented Camera System for Improved Spatial Localization and Spatial Orientation Determination, FSP1700, US Patent 11,132,590, granted 27 August 2021.

15. Brent, R. Delivery and Monitoring Use And Dynamic Adjustment of Licensed Content in Augmented Reality, 2018, FSP1308. *Patent abandoned.*

16. Brent, R. Procedural Language and Content Generation Environment for use in Augmented Reality/Mixed Reality Systems to Support Laboratory and Related Operations. FSP1306c-1. Continuation of US Patent 10,963,750, granted as US Patent 11,386,303, July 2022.

17. Brent, R. Improved Somatic and Somatosensory Guidance in Virtual and Augmented Reality Environments, Filed February 2021, FSP1307. US Patent 11,756,272 issued September 2023.

18. Brent, R. Augmented Camera System for Improved Spatial Localization and Spatial Orientation Determination, FSP1700 c-1, Continuation of US Patent 11,132,590, Filed February 2021. Issued August 2023 as US patent 11,727,238.

19. Engel, S., Peria, W., and Brent, R. Control Structures for Procedural Guidance Systems, FSP 2077, US Patent 12,567,214 issued 3 March 2026